

Advanced Linear Algebra: Foundations to Frontiers

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1.2.1 Absolute value ¶

Remark 1.2.1.1. Don't Panic!

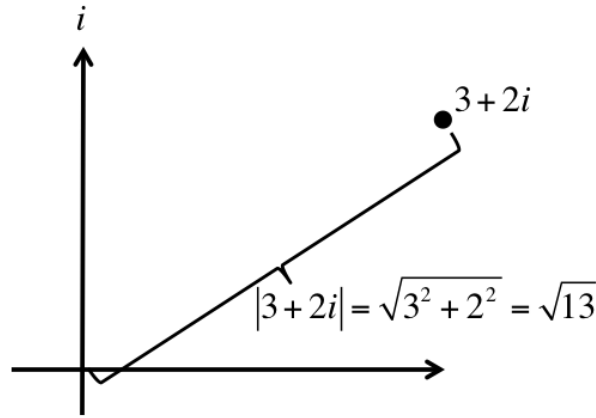
In this course, we mostly allow scalars, vectors, and matrices to be complex-valued. This means we will use terms like "conjugate" and "Hermitian" quite liberally. You will think this is a big deal, but actually, if you just focus on the real case, you will notice that the complex case is just a natural extension of the real case.

fit width

Recall that $|\cdot| : \mathbb{C} \rightarrow \mathbb{R}$ is the function that returns the absolute value of the input. In other words, if $\alpha = \alpha_r + \alpha_c i$, where α_r and α_c are the real and imaginary parts of α , respectively, then

$$|\alpha| = \sqrt{\alpha_r^2 + \alpha_c^2}.$$

The absolute value (magnitude) of a complex number can also be thought of as the (Euclidean) distance from the point in the complex plane to the origin of that plane, as illustrated below for the number $3 + 2i$.



Alternatively, we can compute the absolute value as

$$\begin{aligned}
 |\alpha| &= \\
 &= \sqrt{\alpha_r^2 + \alpha_c^2} \\
 &= \sqrt{\alpha_r^2 - \alpha_c \alpha_r i + \alpha_r \alpha_c i + \alpha_c^2} \\
 &= \sqrt{(\alpha_r - \alpha_c i)(\alpha_r + \alpha_c i)} \\
 &= \sqrt{\bar{\alpha} \alpha},
 \end{aligned}$$

where $\bar{\alpha}$ denotes the complex conjugate of α :

$$\bar{\alpha} = \overline{\alpha_r + \alpha_c i} = \alpha_r - \alpha_c i.$$

The absolute value function has the following properties:

- $\alpha \neq 0 \Rightarrow |\alpha| > 0$ ($|\cdot|$ is positive definite),
- $|\alpha\beta| = |\alpha||\beta|$ ($|\cdot|$ is homogeneous), and
- $|\alpha + \beta| \leq |\alpha| + |\beta|$ ($|\cdot|$ obeys the triangle inequality).

Norms are functions from a domain to the real numbers that are positive definite, homogeneous, and obey the triangle inequality. This makes the absolute value function an example of a norm.

The below exercises help refresh your fluency with complex arithmetic.

Homework 1.2.1.1.

1. $(1 + i)(2 - i) =$
2. $(2 - i)(1 + i) =$
3. $\overline{(1 - i)(2 - i)} =$
4. $\overline{\overline{(1 - i)(2 - i)}} =$
5. $\overline{(2 - i)(1 - i)} =$
6. $(1 - i)\overline{(2 - i)} =$

▼ Solution

1. $(1+i)(2-i) = 2 + 2i - i - i^2 = 2 + i + 1 = 3 + i$
2. $(2-i)(1+i) = 2 - i + 2i - i^2 = 2 + i + 1 = 3 + i$
3. $\overline{(1-i)}(2-i) = (1+i)(2-i) = 2 - i + 2i - i^2 = 3 + i$
4. $\overline{(1-i)}(2-i) = \overline{(1+i)(2-i)} = \overline{2 - i + 2i - i^2} = \overline{2 + i + 1} = \overline{3 + i} = 3 - i$
5. $\overline{(2-i)}(1-i) = (2+i)(1-i) = 2 - 2i + i - i^2 = 2 - i + 1 = 3 - i$
6. $(1-i)\overline{(2-i)} = (1-i)(2+i) = 2 + i - 2i - i^2 = 2 - i + 1 = 3 - i$

Homework 1.2.1.2. Let $\alpha, \beta \in \mathbb{C}$.

1. ALWAYS/SOMETIMES/NEVER: $\alpha\beta = \beta\alpha$.
2. ALWAYS/SOMETIMES/NEVER: $\overline{\alpha}\beta = \overline{\beta\alpha}$.

► [Hint](#) ► [Answer](#) ▼ [Solution](#)

1. ALWAYS: $\alpha\beta = \beta\alpha$.

Proof:

$$\begin{aligned}
 &\alpha\beta \\
 &= \quad < \text{substitute} > \\
 &(\alpha_r + \alpha_c i)(\beta_r + \beta_c i) \\
 &= \quad < \text{multiply out} > \\
 &\alpha_r \beta_r + \alpha_r \beta_c i + \alpha_c \beta_r i - \alpha_c \beta_c \\
 &= \quad < \text{commutativity of real multiplication} > \\
 &\beta_r \alpha_r + \beta_r \alpha_c i + \beta_c \alpha_r i - \beta_c \alpha_c \\
 &= \quad < \text{factor} > \\
 &(\beta_r + \beta_c i)(\alpha_r + \alpha_c i) \\
 &= \quad < \text{substitute} > \\
 &\beta\alpha.
 \end{aligned}$$

2. SOMETIMES: $\overline{\alpha}\beta = \overline{\beta\alpha}$.

An example where it is true: $\alpha = \beta = 0$.

An example where it is false: $\alpha = 1$ and $\beta = i$. Then $\overline{\alpha}\beta = 1 \times i = i$ and $\overline{\beta\alpha} = -i \times 1 = -i$.

Homework 1.2.1.3. Let $\alpha, \beta \in \mathbb{C}$.

ALWAYS/SOMETIMES/NEVER: $\overline{\overline{\alpha}\beta} = \overline{\beta\alpha}$.

Yet another way of presenting the proof uses an "equivalence style proof." The idea is to start with the equivalence you wish to prove correct:

$$\overline{\overline{\alpha}\beta} = \overline{\beta\alpha}$$

and through a sequence of equivalent statements argue that this evaluates to TRUE:

$$\overline{\overline{\alpha}\beta} = \overline{\beta\alpha}$$

$$\Leftrightarrow \quad < \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i >$$

$$\overline{(\alpha_r + \alpha_c i)(\beta_r + \beta_c i)} = \overline{(\beta_r + \beta_c i)(\alpha_r + \alpha_c i)}$$

$$\Leftrightarrow \quad < \text{conjugate} \times 2 >$$

$$\overline{(\alpha_r - \alpha_c i)(\beta_r + \beta_c i)} = \overline{(\beta_r - \beta_c i)(\alpha_r + \alpha_c i)}$$

$$\Leftrightarrow \quad < \text{multiply out} \times 2 >$$

$$\overline{\alpha_r \beta_r + \alpha_r \beta_c i - \alpha_c \beta_r i + \alpha_c \beta_c} = \overline{\beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c}$$

$$\Leftrightarrow \quad < \text{conjugate} >$$

$$\alpha_r \beta_r - \alpha_r \beta_c i + \alpha_c \beta_r i + \alpha_c \beta_c = \beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c$$

$$\Leftrightarrow \quad < \text{subtract equivalent terms from left-hand side and right-hand side} >$$

$$0 = 0$$

$$\Leftrightarrow \quad < \text{algebra} >$$

TRUE.

By transitivity of equivalence, we conclude that $\overline{\overline{\alpha}\beta} = \overline{\beta\alpha}$ is TRUE.

Proofs in mathematical textbooks seem to always be wonderfully smooth arguments that lead from the left-hand side of an equivalence to the right-hand side. In practice, you may want to start on the left-hand side, and apply a few rules:

$$\overline{\overline{\alpha}\beta}$$

$$= \quad < \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i >$$

$$\overline{(\alpha_r + \alpha_c i)(\beta_r + \beta_c i)}$$

$$= \quad < \text{conjugate } \alpha >$$

$$\overline{(\alpha_r - \alpha_c i)(\beta_r + \beta_c i)}$$

$$= \quad < \text{multiply out} >$$

$$\overline{(\alpha_r \beta_r - \alpha_c \beta_r i + \alpha_r \beta_c i + \alpha_c \beta_c)}$$

$$= \quad < \text{conjugate} >$$

$$\alpha_r \beta_r + \alpha_c \beta_r i - \alpha_r \beta_c i + \alpha_c \beta_c$$

and then move on to the right-hand side, applying a few rules:

$$\begin{aligned}
& \overline{\beta\alpha} \\
&= \langle \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i \rangle \\
& \overline{(\beta_r + \beta_c i)(\alpha_r + \alpha_c i)} \\
&= \langle \text{conjugate } \beta \rangle \\
& (\beta_r - \beta_c i)(\alpha_r + \alpha_c i) \\
&= \langle \text{multiply out} \rangle \\
& \beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c.
\end{aligned}$$

At that point, you recognize that

$$\begin{aligned}
& \alpha_r \beta_r + \alpha_c \beta_r i - \alpha_r \beta_c i + \alpha_c \beta_c \\
&= \\
& \beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c
\end{aligned}$$

since the second is a rearrangement of the terms of the first. Optionally, you then go back and presents these insights as a smooth argument that leads from the expression on the left-hand side to the one on the right-hand side:

$$\begin{aligned}
& \overline{\alpha\beta} \\
&= \langle \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i \rangle \\
& \overline{(\alpha_r + \alpha_c i)(\beta_r + \beta_c i)} \\
&= \langle \text{conjugate } \alpha \rangle \\
& \overline{(\alpha_r - \alpha_c i)(\beta_r + \beta_c i)} \\
&= \langle \text{multiply out} \rangle \\
& \overline{(\alpha_r \beta_r - \alpha_c \beta_r i + \alpha_r \beta_c i + \alpha_c \beta_c)} \\
&= \langle \text{conjugate} \rangle \\
& \alpha_r \beta_r + \alpha_c \beta_r i - \alpha_r \beta_c i + \alpha_c \beta_c \\
&= \langle \text{rearrange} \rangle \\
& \beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c \\
&= \langle \text{factor} \rangle \\
& (\beta_r - \beta_c i)(\alpha_r + \alpha_c i) \\
&= \langle \text{definition of conjugation} \rangle \\
& \overline{(\beta_r + \beta_c i)(\alpha_r + \alpha_c i)} \\
&= \langle \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i \rangle \\
& \overline{\beta\alpha}.
\end{aligned}$$

$$\begin{aligned}
& \overline{\alpha\beta} \\
&= \overline{\langle \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i \rangle} \\
&= \overline{(\alpha_r + \alpha_c i)(\beta_r + \beta_c i)} \\
&= \overline{\langle \text{conjugate } \alpha \rangle} \\
&= \overline{(\alpha_r - \alpha_c i)(\beta_r + \beta_c i)} \\
&= \overline{\langle \text{multiply out} \rangle} \\
&= \overline{(\alpha_r \beta_r - \alpha_c \beta_r i + \alpha_r \beta_c i + \alpha_c \beta_c)} \\
&= \overline{\langle \text{conjugate} \rangle} \\
&= \overline{\alpha_r \beta_r + \alpha_c \beta_r i - \alpha_r \beta_c i + \alpha_c \beta_c} \\
&= \overline{\langle \text{rearrange} \rangle} \\
&= \overline{\beta_r \alpha_r + \beta_r \alpha_c i - \beta_c \alpha_r i + \beta_c \alpha_c} \\
&= \overline{\langle \text{factor} \rangle} \\
&= \overline{(\beta_r - \beta_c i)(\alpha_r + \alpha_c i)} \\
&= \overline{\langle \text{definition of conjugation} \rangle} \\
&= \overline{(\beta_r + \beta_c i)(\alpha_r + \alpha_c i)} \\
&= \overline{\langle \alpha = \alpha_r + \alpha_c i; \beta = \beta_r + \beta_c i \rangle} \\
&= \overline{\beta\alpha}
\end{aligned}$$

Homework 1.2.1.4. Let $\alpha \in \mathbb{C}$.

ALWAYS/SOMETIMES/NEVER: $\overline{\alpha}\alpha \in \mathbb{R}$

▼ Answer

▼ Solution

Let $\alpha = \alpha_r + \alpha_c i$. Then

$$\begin{aligned}
& \overline{\alpha}\alpha \\
&= \overline{\langle \text{instantiate} \rangle} \\
&= \overline{(\alpha_r + \alpha_c i)(\alpha_r + \alpha_c i)} \\
&= \overline{\langle \text{conjugate} \rangle} \\
&= \overline{(\alpha_r - \alpha_c i)(\alpha_r + \alpha_c i)} \\
&= \overline{\langle \text{multiply out} \rangle} \\
&= \overline{\alpha_r^2 + \alpha_c^2},
\end{aligned}$$

which is a real number.

ALWAYS.

Now prove it!

Homework 1.2.1.5. Prove that the absolute value function is homogeneous:

$|\alpha\beta| = |\alpha||\beta|$ for all $\alpha, \beta \in \mathbb{C}$.

▼ Solution

$$\begin{aligned}
 |\alpha\beta| &= |\alpha||\beta| \\
 \Leftrightarrow &< \text{squaring both sides simplifies} > \\
 |\alpha\beta|^2 &= |\alpha|^2|\beta|^2 \\
 \Leftrightarrow &< \text{instantiate} > \\
 |(\alpha_r + \alpha_c i)(\beta_r + \beta_c i)|^2 &= |\alpha_r + \alpha_c i|^2 |\beta_r + \beta_c i|^2 \\
 \Leftrightarrow &< \text{algebra} > \\
 |(\alpha_r \beta_r - \alpha_c \beta_c) + (\alpha_r \beta_c + \alpha_c \beta_r)i|^2 &= (\alpha_r^2 + \alpha_c^2)(\beta_r^2 + \beta_c^2) \\
 \Leftrightarrow &< \text{algebra} > \\
 (\alpha_r \beta_r - \alpha_c \beta_c)^2 + (\alpha_r \beta_c + \alpha_c \beta_r)^2 &= (\alpha_r^2 + \alpha_c^2)(\beta_r^2 + \beta_c^2) \\
 \Leftrightarrow &< \text{algebra} > \\
 \alpha_r^2 \beta_r^2 - 2\alpha_r \alpha_c \beta_r \beta_c + \alpha_c^2 \beta_c^2 + \alpha_r^2 \beta_c^2 + 2\alpha_r \alpha_c \beta_r \beta_c + \alpha_c^2 \beta_r^2 \\
 &= \alpha_r^2 \beta_r^2 + \alpha_r^2 \beta_c^2 + \alpha_c^2 \beta_r^2 + \alpha_c^2 \beta_c^2 \\
 \Leftrightarrow &< \text{subtract equivalent terms from both sides} > \\
 0 &= 0 \\
 \Leftrightarrow &< \text{algebra} > \\
 T
 \end{aligned}$$

Homework 1.2.1.6. Let $\alpha \in \mathbb{C}$.

ALWAYS/SOMETIMES/NEVER: $|\bar{\alpha}| = |\alpha|$.

▼ Answer

▼ Solution

Let $\alpha = \alpha_r + \alpha_c i$.

$$\begin{aligned}
 |\bar{\alpha}| &= < \text{instantiate} > \\
 |\overline{\alpha_r + \alpha_c i}| &= < \text{conjugate} > \\
 |\alpha_r - \alpha_c i| &= < \text{definition of } |\cdot| > \\
 \sqrt{\alpha_r^2 + \alpha_c^2} &= < \text{definition of } |\cdot| > \\
 |\alpha_r + \alpha_c i| &= < \text{instantiate} > \\
 |\alpha|
 \end{aligned}$$

ALWAYS

Now prove it!

